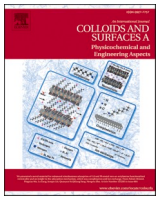
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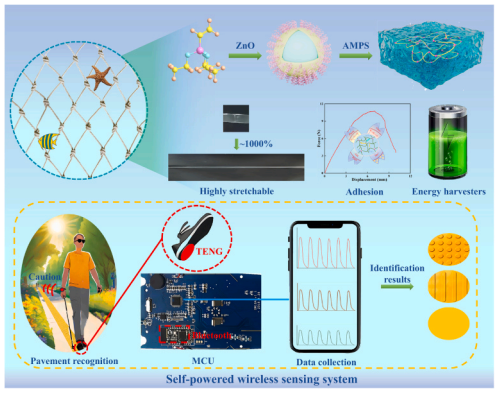
Fishnet-inspired dual-crosslinked nanocomposite hydrogels with core-shell SiO_2@ZnO for self-powered sensing and intelligent mobility assistance systems

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GRAPHICAL ABSTRACT



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ABSTRACT

Enhancing the toughness, conductivity and adhesion of hydrogels while reducing hysteresis is crucial for the development of flexible hydrogel sensors. Inspired by fishing nets, silica-coated zinc oxide core-shell nanoparticles (SiO_2@ZnO) with unsaturated double bonds on their surface were synthesized through an in-situ sol-gel method. These nanoparticles were incorporated into 2-acrylamido-2-methylpropane sulfonic acid (AMPS), forming a double-crosslinked, nano-SiO_2 reinforced multifunctional composite hydrogel (SiO_2@ZnO/PAMPS, PAS) through radical polymerization. The results show that the hydrogel has a low hysteresis rate of less than 10 % at a breaking strain of 1412 %, and the adhesion strength with pig skin, which simulates human skin, reaches 34.79 kPa. Additionally, strain sensors based on this hydrogel show a high sensitivity value of 1.26 and a fast response time (214 ms) within a strain range of 200–700 %. Furthermore, a triboelectric nanogenerator (TENG) based on PAS demonstrates an open-circuit voltage of 131.2 V and a power density of 239.13 mW/m^2. This TENG can function not only as a power source for electronic devices but also detect differences signal

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generated while walking on surfaces with different texture. By integrating it with a Bluetooth system, a self-powered intelligent pavement wireless sensing system aimed at aiding the mobility of the visually impaired has been developed, highlighting the potential application of PAS hydrogel sensors in self-powered sensing fields.

1. Introduction

Flexible wearable sensors have gained extensive attention for their ability to mimic the flexibility [1], softness [2] and sensory features [3] of human skin, they are crucial in applications such as intelligent electronic skin [4], human motion monitoring [5,6], soft robots [7] and human-computer interaction [8]. The challenge lies in designing soft and deformable conductive materials [9] that can convert external mechanical energy into electrical signals [10,11]. Hydrogels are widely used in the manufacturing of strain and pressure sensors for human motion monitoring due to their flexibility [12,13], tunable conductivity [14], and biocompatibility [15,16]. In practical applications, hydrogels sensors must accurately detect various types of deformation and pressure from different movements [17,18], necessitating specific performance. Firstly, hydrogels must demonstrate high stretchability to withstand mechanical forces without damage [19,20]. Secondly, the hydrogel sensors should have low hysteresis during multiple cyclic stretching or compression to minimize energy dissipation, ensuring reliability and durability [21–25]. Finally, hydrogels should exhibit strong adhesion to various substrates to effectively sense strain and pressure [26–29].

Researchers have pursued various strategies to enhance the mechanical properties of hydrogels. For example, by introducing metal ions into the chemically cross-linked polyacrylic acid (PAA) and chitosan (CS) hydrogel, a physical cross-linking structure is formed with the carboxyl groups in PAA and the charged ammonium ions in CS [30]. This results in the hydrogel with a double cross-linked network structure, which has excellent stretching properties and a elongation at break up to 1080 %. Additionally, nanocomposite hydrogels with functionalized zeolite imidazolate framework 71 and polydioxyethiophene manifested an elongation at break exceeding 2200 % due to the existence of core-shell structure nanoparticles [31]. Moreover, a series of rotane crosslinkers with various molecular weights were synthesized by combining γ-cyclodextrin and polyethylene glycol diacrylate. These crosslinkers were co-polymerized with monomers to create slip-ring hydrogels that achieved a remarkable breaking elongation of 6467 % [32]. Despite these advancements, issues of lag and energy dissipation persist during stretching or compression. However, incorporation of "entanglement" within the hydrogel can help reduce stress concentration and lower hysteresis rates [33–36]. For example, hydrogel with Mxene as an entanglement can efficaciously transfer stress to adjacent molecular chains, circumventing stress concentration, thereby giving rise to a lag rate of less than 7 % and merely 900 % elongation at break [37].

The adhesive properties of hydrogels are crucial for their application as flexible wearable sensors. Adhesion of hydrogels can be enhanced via interaction forces such as hydrogen bonding [38], ionic bonding [39], and mechanical interlocking [40,41]. The amphoteric ions and polar groups in the monomer [2-(methylacryloxy) ethyl] dimethyl-(3-propyl sulfonate) ammonium hydroxide can significantly improve adhesion to multiple substances through hydrogen bonding and ion association [42]. Nevertheless, enhancing mechanical properties, reducing hysteresis and increasing adhesion often present conflicting challenges. High strechability performance typically requires hydrogels to have a higher covalent cross-linking density and energy dissipation to suppress elastic shrinkage and enhance the viscous flow within the polymer chains [43–45], while low hysteresis demands that hydrogels reduce the interaction of sacrificial bonds (reversible covalent bonds and non-covalent interactions) to decrease the energy dissipation [46–49]. Moreover, hydrogels with high strechability and low hysteresis often struggle with weak adhesion strength [50,51]. Therefore, there are still limitations in developing hydrogels with excellent comprehensive properties.

Inspired by fishing nets, this work synthesized core-shell structured particles of silica-coated zinc oxide nanoparticles (SiO_2@ZnO) with unsaturated double bonds on their surface using an in-situ sol-gel approach. These particles are introduced as inorganic nano "entanglement points" into the cross-linked network structure of hydrogels. Moreover, The unsaturated double bonds on the surface of the core-shell SiO_2@ZnO nanoparticles provide reactive sites for covalent crosslinking, enhancing the density of the hydrogels' crosslinking network. Additionally, the Zn^{2+} in the core of SiO_2@ZnO nanoparticles can increase ion concentration to enhance the electrical conductivity while also forming an ion cross-linked network structure, which contributes to enhanced toughness. Herein, the SiO_2@ZnO nanoparticle with unsaturated groups and core-shell structure was introduced into 2-acrylamide-2-methylpropanesulfonic acid (AMPS) to synthesize a composite hydrogel (PAS) with a fishnet-like structure, featuring high stretchability, low hysteresis, and high conductivity through radical polymerization. Furthermore, the presence of sulfonic acid groups within PAMPS confers upon PAS hydrogels the properties of excellent adhesion and biocompatibility. With these advantages, the PAS hydrogel can serve as a sensor for detecting human motion and can also be assembled into a triboelectric nanogenerator (TENG) for harvesting mechanical energy. More importantly, integrating the PAS-TENG with a Bluetooth system can create a self-powered intelligent pavement sensing system for assisting visually impaired individuals. The findings suggest that outcomes manifest the potential of the PAS hydrogel sensors have promising applications in self-powered wireless sensing and related fields.

2. Experimental section

2.1. Materials

Acrylamido-2-methylpropanesulfonic acid (AMPS) was procured from Sigma Aldrich. N,N′-Methylenebisacrylamide (MBA) was obtained from Beijing Inokai Technology Co. Zinc oxide (ZnO) nanoparticles were sourced from McLean Shanghai. Vinyltriethoxysilane (VTES) and ammonium persulfate (APS) were acquired from Aladdin (Shanghai) Company. All chemical reagents were not subjected to further purification prior to use.

2.2. Synthesis of SiO_2@ZnO nanoparticles

First, 0.75 mL of VTES was pipetted into 100 mL of anhydrous ethanol and stirred at room temperature for 1 hour. Meanwhile, 3.5 g of zinc oxide nanoparticles were weighed and added to a mixture of 5 mL of ammonia and 20 mL of deionized water, then ultrasonically dispersed. After completely dispersion, this solution was added to the ethanol containing VTES, and the mixture was stirred continuously at 45 °C for 24 h to ensure completed dehydration condensation. The resulting solution was then subjected to centrifugal washing and dried under vacuum at 60 °C for 12 h to yield SiO_2@ZnO.

2.3. Preparation of PAS hydrogels

The hydrogels were prepared using a "one-pot method". Firstly, 3 g of AMPS, 0.03 g of MBA, 0.02 g of APS and varying amounts of SiO_2@ZnO (0 wt%, 0.5 wt%, 1 wt%, 1.5 wt%, and 2 wt%) were simultaneously added to 2.5 mL of deionized water and magnetically stirred for 1 h. The mixture was then ultrasonicated for 5 min to remove internal air

bubbles. Finally, the homogenized solution was poured into a petri dish and placed in an oven at 80 °C for 8 h, resulting in a double-crosslinked, nano-SiO_2 reinforced multifunctional composite hydrogel (SiO_2@ZnO/PAMPS, PASx), where x indicates the SiO_2@ZnO mass fraction.

2.4. Preparation of PAS hydrogels-based TENG

Mix components A and B of Ecoflex 00–30 silicone rubber in a 1:1 ratio, pour into molds, and cure for 6 h. Once cured, cut the hydrogel to size, place it in the molds, insert pre-constructed wires, and cover with an equal amount of silicone rubber. After full curing, a "sandwich" structure of PAS-TENG (2.5 × 2.5 × 0.3 cm^3) will be achieved.

2.5. Characterization

Fourier transform infrared spectroscopy (Nicolet Nexus 470) was used to analyze functional groups interaction. Morphology was observed via transmission electron microscopy (JEOL-JEM-2100F), while transparency was assessed with a UV–visible spectrophotometer (UV-9000S), The mechanical properties of the hydrogels were investigated with a universal testing machine (UTM4503BLXY), where the samples were cut into a dumbbell shape with a tensile rate of 50 mm/min; shear tensile tests were conducted to assess adhesion properties of the hydrogels. The conductivity of the hydrogels and the electrostatic output of the hydrogel-based TENG were measured using a programmable electrostatic meter (Keithley 6514). The pressure applied to the TENG was recorded by a pressure tester (M5-20).

3. Results and discussion

3.1. Design principle and fabrication of PAS hydrogels

This paper presents the designed and synthesis of a multifunctional hydrogel (PAS) was, composed of AMPS as the monomer, APS as the initiator, and MBA and SiO_2@ZnO as cross-linkers. As shown in Fig. 1a and Fig. S1, the synthesis of SiO_2@ZnO begins with the hydrolysis of alkoxyl groups of VTES to silanol groups, which then condense to form branched polysiloxanes. Subsequently, ZnO is added, allowing the condensed oligomer to adsorb onto the hydroxyl group on the surface of ZnO through hydrogen bonding. This result in core-shell structured particles (SiO_2@ZnO) with unsaturated double bonds, capable of forming covalent network structure with monomers containing double bonds. As shown in Fig. S2, the SiO_2@ZnO particles possess a distinct core-shell structure, with a 0.51 nm lattice spacing in the inner core corresponding to the (001) facet of Zn^{2+}, and silica-rich outer shell consisting unsaturated double bonds. According to the particle size analysis results (Fig. S3), the initial ZnO particles had an average diameter of approximately 500 nm. After modification with vinyl triethoxysilane to form a core-shell structured SiO_2@ZnO, the particle size distribution exhibited a significant increase. This suggests that most of the ZnO nanoparticles successfully formed core-shell structures coated with SiO_2. Furthermore, zeta potential analysis confirms that ZnO nanoparticles carry a positive charge, while SiO_2@ZnO particles display a negative charge with an absolute value greater than 30 mV. This result strongly supports the conclusion that vinyl triethoxysilane has effectively modified the ZnO surface, ensuring its long-term stability. To further confirm the successful synthesis of SiO_2@ZnO, Fourier transform

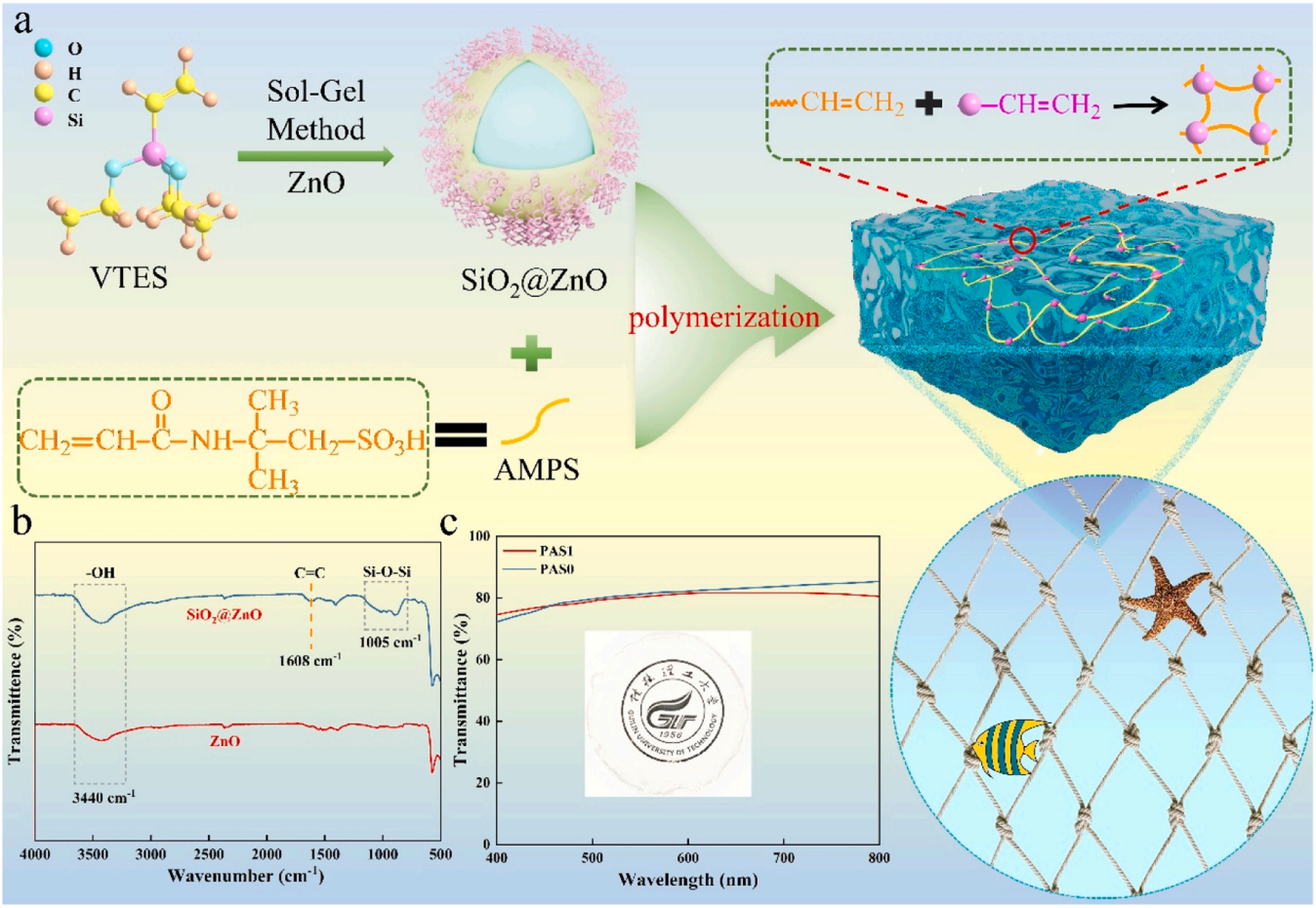


Fig. 1. a) Preparation process of PAS hydrogel. b) FTIR image of SiO_2@ZnO and ZnO. c) Transparency of PAS hydrogel.

infrared spectroscopy (FTIR) was conducted to verify the functional group interactions. As shown in Fig. 1b, the stretching vibrational peaks of -OH are all at 3440 cm^{-1} and new absorption peaks at 1608 cm^{-1} and 1005 cm^{-1} for C═C stretching vibration and Si-O-Si antisymmetric stretching vibration, respectively, which indicated that the VTES was successfully encapsulated on the surface of ZnO. The PAS hydrogel was fabricated through a one-pot thermal polymerization method, where the monomer, initiator and cross-linking agent are dissolved in deionized water, stirred to homogeneous, and reacted at 80 °C for 8 h. The formation results from multiple interactions among the components, where thermal initiation breaks the double bonds on AMPS and SiO_2@ZnO, allowing them to re-covalently graft through radical polymerization,

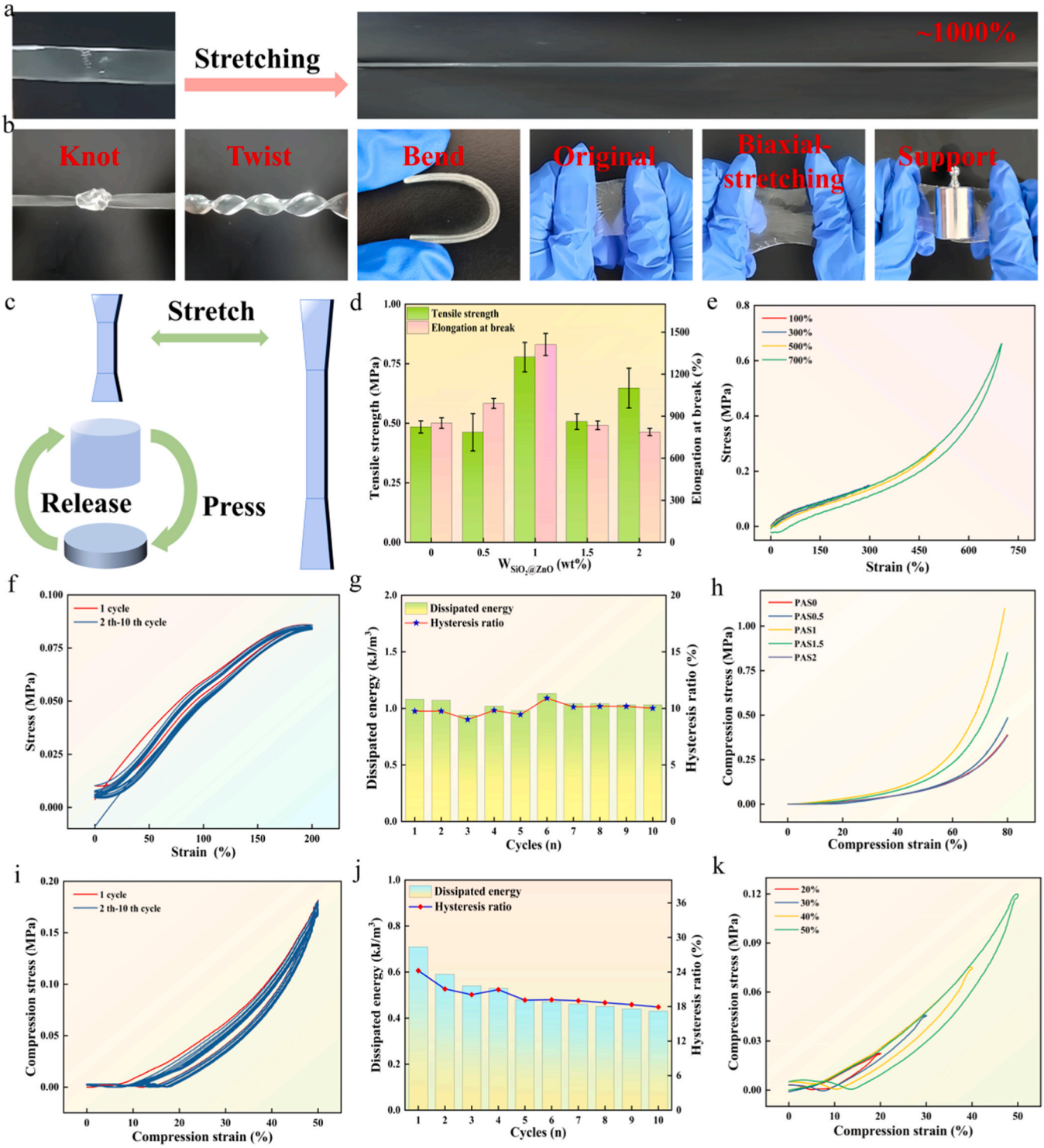


Fig. 2. Optical images of PAS1 hydrogels: a) Stretching, b) Knotting, twisting, bending, bidirectional stretching, and load bearing. c) Schematic illustration of hydrogel stretching and compression. d) Tensile strength and elongation at break of the hydrogel. e) Cycling curves of PAS1 hydrogel at various tensile strains. f) Ten tensile cycles of PAS1 hydrogel at 200 % strain. g) Dissipation energy and hysteresis rate of PAS1 hydrogel for 10 tensile cycles at 200 % strain. h) Compressive strength of the hydrogel. i) 10 compression cycles of PAS1 hydrogel at 80 % strain. j) Dissipation energy and hysteresis rate of PAS1 hydrogel for 10 compression cycles at 80 % strain. k) Cycling curves of PAS1 hydrogel at different compressive strains.

creating interwoven polymer chains. These elongated chains are intricately interwoven with each other, and SiO_2 particles serve as "entanglement" in fishing nets, thereby bringing about resilient PAS hydrogels. In addition, the internal morphology of the hydrogel was systematically characterized using scanning electron microscopy (SEM). As shown in Fig. S4 that the hydrogel without SiO_2@ZnO exhibited a smooth internal structure devoid of discernible particles. By contrast, the hydrogel containing 1 wt% SiO_2@ZnO featured distinct particles embedded within its matrix, which acted as "nodes" in a fishnet-like structure, thereby significantly enhancing the structural integrity of the hydrogel. Importantly, the SiO_2@ZnO nanoparticles were homogeneously dispersed throughout the hydrogel matrix, with no evidence of significant agglomeration. Furthermore, the particle sizes were found to be highly consistent. Additionally, Fig. 1c shows that incorporating SiO_2@ZnO does not affect the transparency of the hydrogel, reflecting structural furthermore, the numerous hydrogen bonds and ionic interactions in the PAS hydrogels endow them self-healing and strong adhesion properties.

3.2. Mechanical properties

PAS hydrogels exhibit outstanding mechanical properties due to the "entanglement" enhancement of SiO_2 nanoparticles alongside intermolecular covalent and non-covalent interactions. As shown in Fig. 2a, the PAS hydrogel can stretch up to approximately 1000 % without fracture. Additionally, it can be deformed in multiple manners at will, such as being knotted, twisted, and bent, and it demonstrates a certain degree of supporting ability, withstanding a load of 200 g after being stretched in both directions (Fig. 2b). These results indicate that the PAS hydrogels possesse favorable mechanical. In order to further understand the influence of SiO_2@ZnO on the these properties, the tensile properties of PAS hydrogels with varying SiO_2@ZnO contents were measured. As shown in Fig. 2d, the mechanical properties initially increase with SiO_2@ZnO content, peaking at 1 wt%—yielding a tensile strength and elongation at break were 0.78 MPa and 1412 %, respectively. This is attributed to SiO_2@ZnO serves as a "entanglement" that enhance covalent and non-covalent interactions within the polymer network, and chain entanglement takes place, thus increasing cross-linking density, and promoting energy dissipation, and relieving stress. However, further increases in SiO_2@ZnO content led to a decline in mechanical properties due to excessive SiO_2@ZnO particles presence hindering the movement of the hydrogel network. In addition to mechanical properties, the reusability and fatigue resistance of hydrogels are crucial for wearable electronics. To assess the stability of mechanical properties during long-term cyclic deformation, PAS1 hydrogels were selected to investigate. As shown in Fig. 2e, cyclic tensile curves exhibited minimal change in hysteresis between 100 % and 700 % strain, likely due to the rapid dissociation and reformation of dynamic hydrogen and reversible ligand bonds within the hydrogel network (Fig. S5a). In a tensile cycling test with a constant 200 % strain (Fig. 2f), the PAS1 hydrogel exhibited a fine hysteresis return line, attributed to energy dissipation from internal friction in the molecular chains in the hydrogel network during the stretching. Remarkably, after 10 cycles, the dissipated energy and hysteresis rate were essentially identical to the original state (Fig. 2g), confirming reliable stability and good fatigue resistance of the PAS1 hydrogel.

Furthermore, the hydrogel exhibits excellent elastic properties. As indicated by the compressive stress-strain curves (Fig. 2h), the compressive strength of the PAS1 hydrogel shows a similar tendency to the tensile strength and likewise reaches a maximum compressive strength of 1.1 MPa at a SiO_2@ZnO content of 1 wt%. Under 50 % strain, 10 consecutive compression cycle tests revealed that the first cycle showed highest hysteresis, with energy dissipation and hysteresis rate decreasing in the subsequent cycles, suggesting that the hydrogel established a recoverable dynamic network during the cycling process (Fig. 2i, j). Compression tests at various strains (20–50 %) (Fig. 2k and Fig. S5b) also demonstrated increase in dissipated energy from 0.05 kJ/m^3 to 0.56 kJ/m^3, indicating the hydrogel can withstand large compressive deformations and recover to its original state, which further indicates it possesses excellent elasticity. In conclusion, the PAS hydrogel possesses outstanding mechanical properties, making it ideal for flexible wearable electronic devices.

3.3. Adhesion behavior

Strong adhesion is crucial for wearable electronics to maintain skin contact during movement, ensuring accurate and reliable signal acquisition. As shown in Fig. 3a, PAS1 hydrogels exhibit strong adhesion to various substrates including rubber, metal, plastic, wood, silicon. Adhesion properties were quantitatively evaluated by sandwiching the PAS1 hydrogel ($1.5 \times 1.5 \times 0.2$ cm^3) between two adherends for lap-shear experiments (Fig. 3b-c). The adhesion strengths of the PAS1 hydrogel to metal, glass, pigskin, plastic, rubber, and cardboard reached 8.49 kPa, 73.70 kPa, 36.45 kPa, 22.21 kPa, 62.58 kPa and 139.09 kPa, respectively (Fig. 3d). This strong adhesion also showed good reproducibility and durability due to reversible physical interactions between the surface of each substrate and the PAS1 hydrogel. Therefore, the adhesion strengths showed only minor decay after each cycle (Fig. 3e). Furthermore, the PAS1 hydrogel adhered well to human skin (Fig. S6), did not fall off with joint bending movements, and did not leave traces on skin after undergoing multiple peelings (Fig. 3f). The excellent adhesion of the hydrogel can be attributed to the abundant dynamic non-covalent interactions with substrates, and the existence of a large number of polar groups (sulfonic acid, hydroxyl) in the hydrogel could offer active sites to form reversible hydrogen and coordination bonds with various substrates (Fig. 3g).

3.4. PAS-based strain sensors

The key index for the application of hydrogels in flexible sensing is to possess good electrical conductivity. SiO_2@ZnO enhances both the mechanical properties and conductivity of hydrogels. First, the impedance of the hydrogel was quantitatively characterized using electrochemical impedance spectroscopy (EIS). Specifically, the Nyquist plot displayed the intercept on the real axis, which directly corresponded to the resistance of the hydrogel (Fig. S7). Thereafter, the conductivity (σ) of the hydrogel was calculated using the formula $\sigma = L/(R \times S)$, as clearly shown in Figure a. The conductivity of the PAS hydrogel attains the optimum value of 1.83 S/m at 1 wt% SiO_2@ZnO mass fraction, due to the increase Zn^{2+} concentration that accelerates ions movement rate in the hydrogel. However, a high concentration of SiO_2@ZnO in the PAS hydrogel leads to a higher cross-linking density, which seriously impedes the ion transport channels. Moreover, the negative effect of this structure is much greater than the positive effect of increasing the ion concentration. Therefore, as the content of SiO_2@ZnO higher than that of at 1 wt%, the conductivity of the PAS hydrogel began to decline. Additionally, the hydrogel will inevitably sustain damage; thus, the hydrogel must also possess a certain self-healing ability. The PAS1 hydrogel, cut in half, can rejoin and withstand a certain amount of tensile force after re-connection (Fig. S8). The healing state of the PAS1 hydrogel can be observed using the 3D profilometer (Fig. S9). In order to quantitatively evaluate the electrical self-healing, the real-time resistance measurements during the repair process of the PAS1 hydrogel was recorded (Fig. 4b). When the PAS1 hydrogel was completely cut off, the circuit was interrupted. However, after the two separated gels were re-contacted, the resistance would quickly return to the initial value, indicating that the PAS1 hydrogel has excellent electrical self-healing performance. Visually, the self-healing PAS1 hydrogel can reignite a bulb with the same luminance (Fig. 4c). The sensitivity of the PAS1 hydrogel to strain was evaluated by analyzing the relative resistance change of PAS1 hydrogel during tensile deformation was recorded ($\Delta R/R_0$). The strain sensitivity (GF) is the ratio of $\Delta R/R_0$ to the strain of the

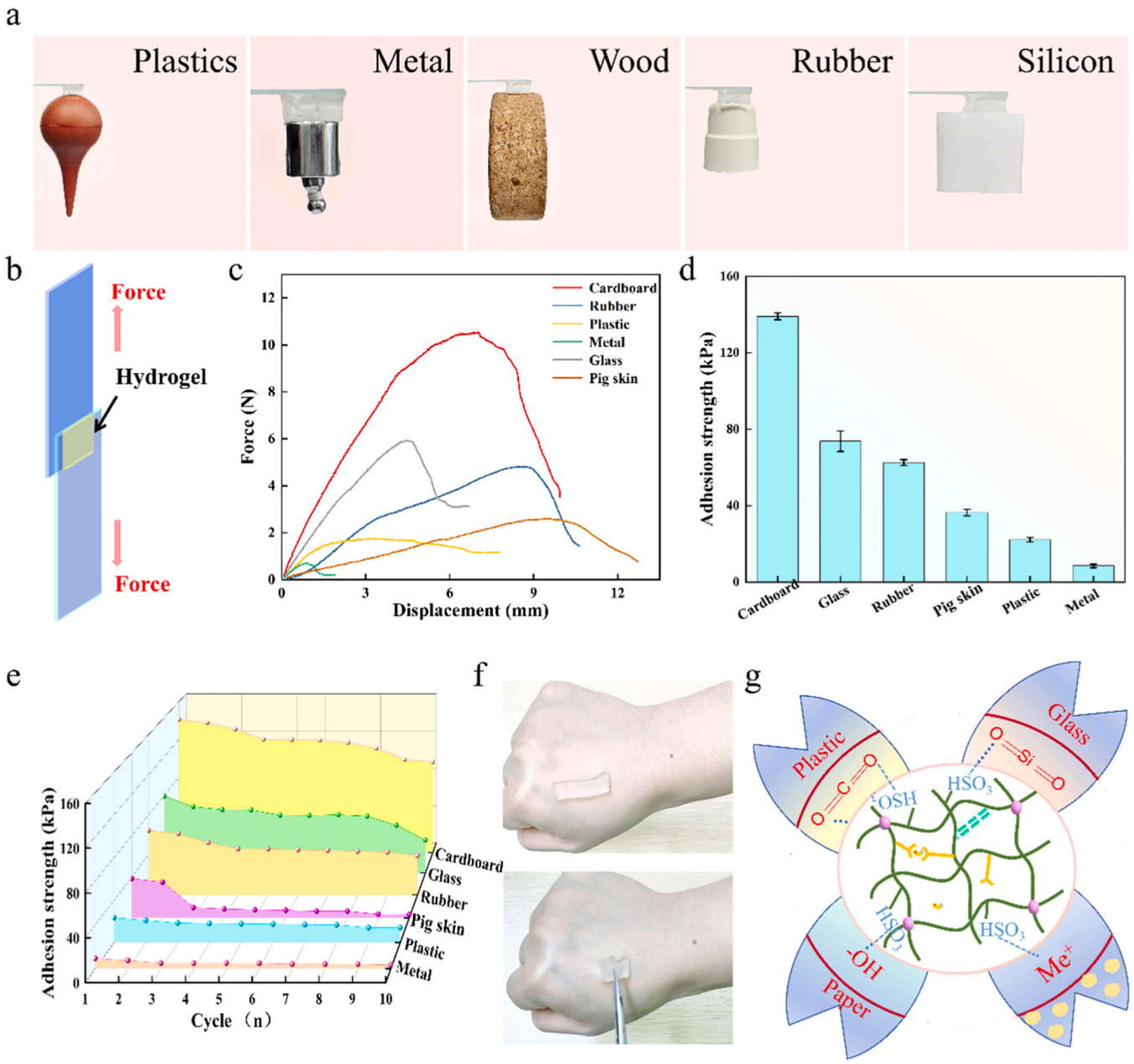


Fig. 3. a) Photographs depicting the adhesion of PAS1 hydrogel to various substrates. b) Schematic illustration of lap-shear test. c) Lap-shear curves of PAS1 hydrogels with diverse substrates. d) Adhesion strength between PAS1 hydrogel and various substrates. e) Repeatable adhesion test of PAS1 hydrogel on various substrates. f) Adhesion of PAS1 hydrogel to human skin with no residue after removal. g) Interaction between PAS1 hydrogel and various substrates.

hydrogel. As shown in Fig. 4d, the GF is 0.49 in the stain range of 10–200 % and 1.26 in the range of 200–700 %, indicating that the PAS1 hydrogel has excellent sensitivity and a wide monitoring range. Fig. 4e and Fig. S10 reveal a curvilinear relationship between tensile strains and relative resistance changes, yielding recognizable, stable, and reproducible signals across various tensile strains. Additionally, the PAS1 hydrogel exhibits a fast response time of 218 ms (Fig. 4f). Owing to its excellent mechanical properties, strong adhesion, high sensitivity, and rapid response time, the PAS1 hydrogel demonstrates significant advantages compared to prior studies (Fig. S11). It is capable of functioning as a strain sensor and can be directly adhered to the skin over joints for the detection of various physiological activities of the human body. As shown in Fig. 4g, the sensor rapidly and accurately detects finger angle changes. It also performed well when the PAS1 hydrogel was adhered to the wrist, elbow and knee (Fig. 4h–j), showing consistent signal changes, and good stability. These results indicate outstanding sensing performance of the PAS1 hydrogel, highlighting its potential application for flexible wearable electronic devices.

3.5. Output performance of PAS-based TENG

TENG is a promising energy harvester that effectively converts mechanical energy into electrical energy [52]. Fig. 5a shows a schematic diagram of a PAS-based single-electrode triboelectric nanogenerator (PAS-TENG), where silicon rubber serves as the negative friction layer, and the PAS hydrogel is encapsulated within it, allowing the device to withstand stretching, bending, and twisting (Fig. 5b). The working principle relies on the coupling effect of friction charging and electrostatic induction [53]. Initially, nitrile rubber (NBR) and silicone rubber generate equal and opposite charges on their surfaces when in contact (Fig. 5c(i)). Upon separation, positive charges in the hydrogel are attracted to the silicone rubber layer to balance the charge (Fig. 5c(ii)).

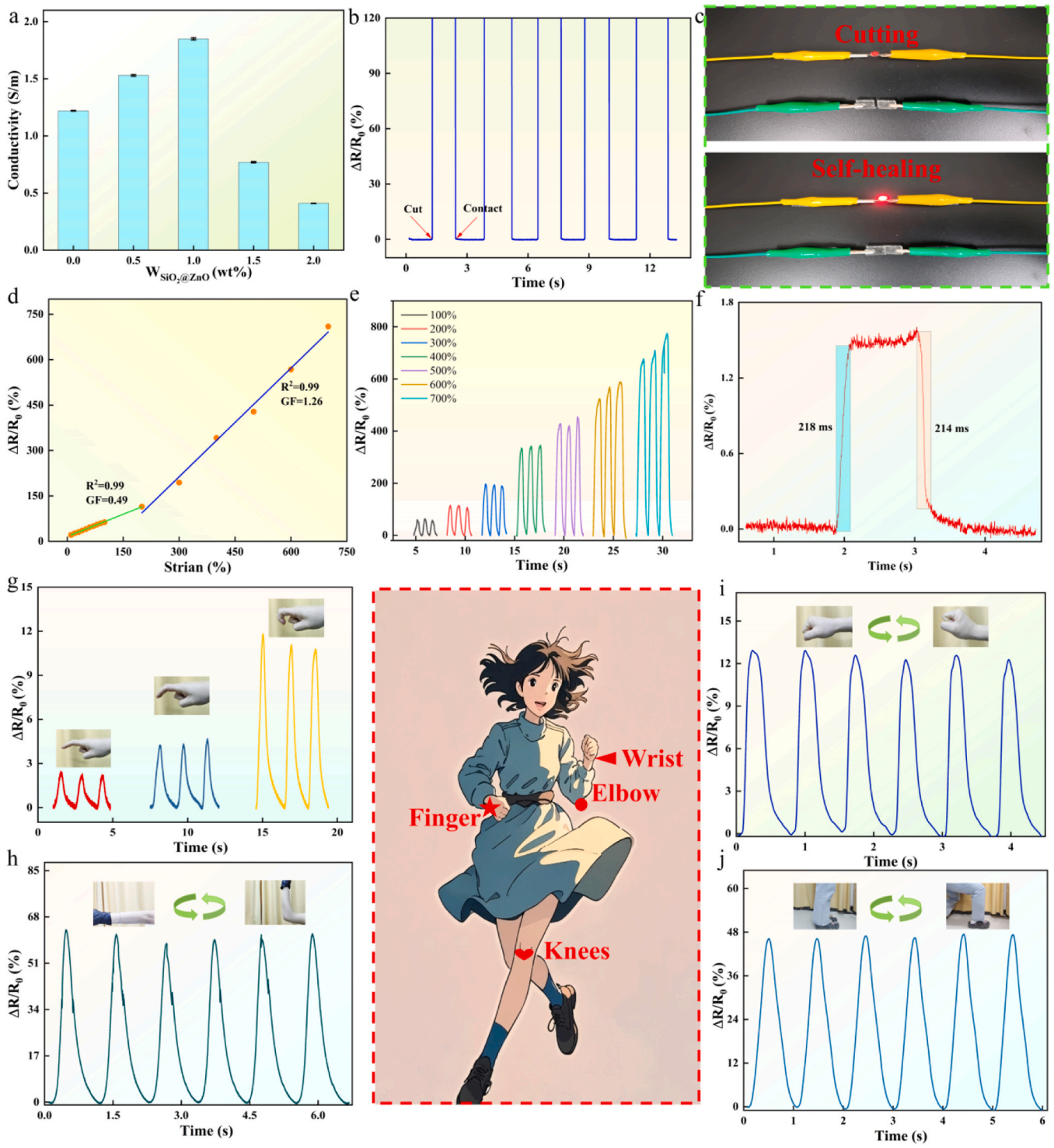


Fig. 4. a) Electrical conductivity of hydrogels. b) Real-time resistance of PAS1 hydrogel during cutoff and self-healing. c) Photograph of self-healing PAS1 hydrogel as a loop to light up the LED. d) GF value of PAS1 hydrogel. e) Relationship between $\Delta R/R_0$ and strain (100–700 %) for PAS1 hydrogel. f) Response time of PAS1 hydrogel. g-j) PAS1 hydrogel for detecting the motion of each joint (finger, wrist, elbow and knee) of human body.

Electrostatic induction, causes electrons to flow through an external circuit, creating current until the positive charge in the hydrogel neutralized the negative charge on the silicone rubber (Fig. 5c(iii)). When the NBR contact silicone rubber again, electrons flow back from the ground to the hydrogel, generating reverse current (Fig. 5c(vi)). This contact-separation motion allows PAS-TENG to effectively convert mechanical energy into electrical energy.

To investigate the influence of SiO_2@ZnO content on the output performance of PAS-TENG, open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), and transferred charge (Q_{sc}) of PAS-TENG were measured at 1 Hz. As shown in Fig. 5d-e and Fig. S12, at a 0.5 wt% SiO_2@ZnO content, the V_{oc}, I_{sc} and Q_{sc} of PAS-TENG were 131.2 V, 1.35 μA, and 40.8 nC, respectively. Further increasing the SiO_2@ZnO content degraded the output performance of PAS-TENG, which may be attributed to the SiO_2@ZnO content causes impact on the dielectric constant of PAS hydrogels. The output properties of PAS1-TENG at different

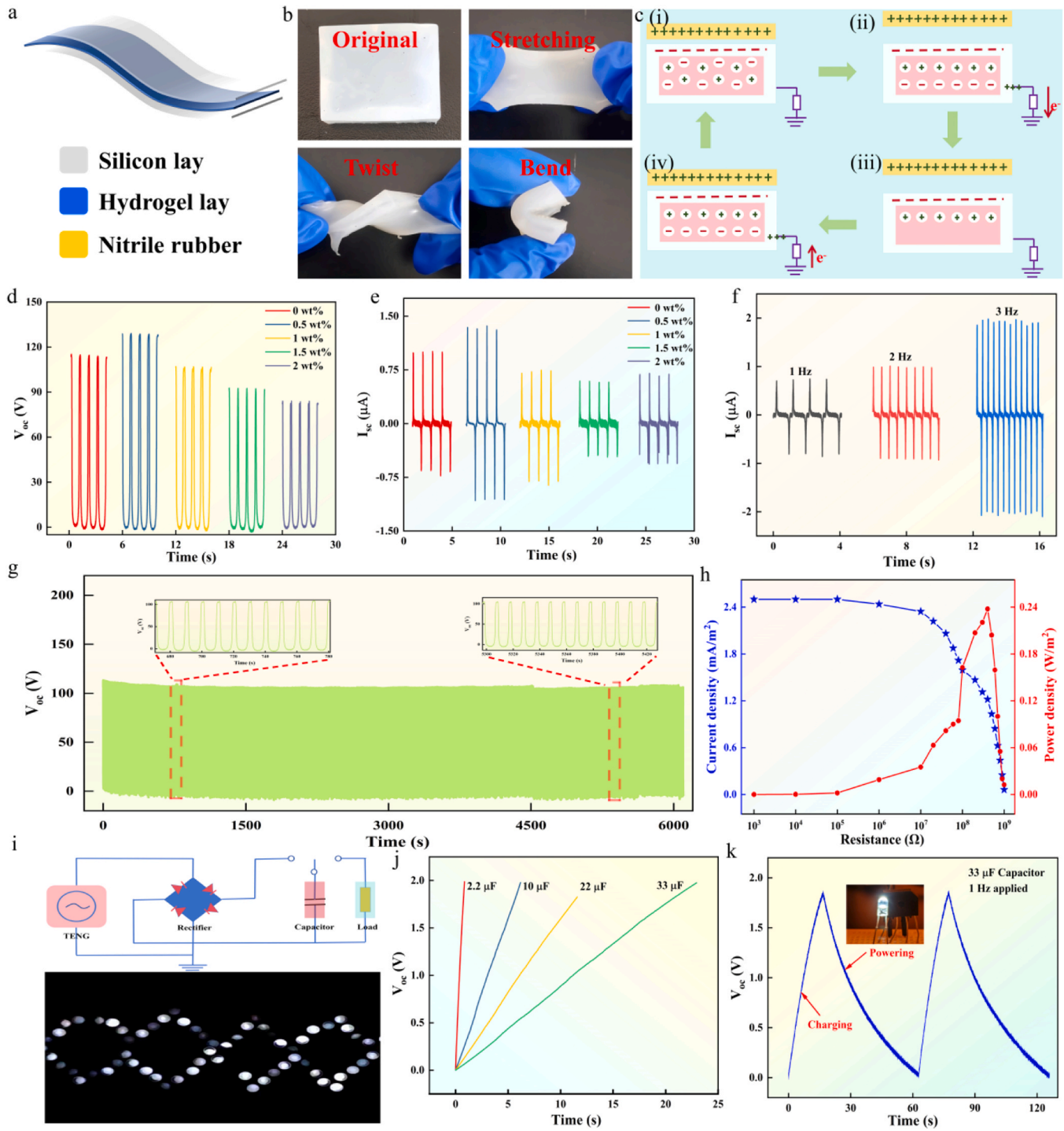


Fig. 5. a) Schematic structure of PAS-TENG. b) Stretching, bending, and twisting states of PAS-TENG. c) Working principle of PAS-TENG. d-e) Open-circuit voltage (V_{oc}) and short-circuit current (I_{sc}) of PAS-TENG. f) Short-circuit current (I_{sc}) of PAS1-TENG at diverse frequencies. g) Output stability of PAS1-TENG for approximately 6000 contact-detachment experiments. h) Power density of PAS1-TENG. i) Circuit configuration with an external load. j) Charging of capacitors with different capacities by PAS1-TENG. k) Charging and discharging process of the PAS1-TENG.

frequencies were also investigated (Fig. 5f and Fig. S13), revealing that the frequency does not affect V_{oc} and Q_{sc}, but I_{sc} increases with the increase of the frequency, which is attributed to the fact that the movement of the electrons is faster at higher frequencies. Additionally, Fig. 5g, indicates that PAS1-TENG maintains stability after approximately 6000 contact-detachment cycles for V_{oc}, showing reliable stability and fatigue resistance.

As shown in Fig. 5h, to evaluate the energy generation capability of the PAS1-TENG, it is connected to an external resistor, and the output current is measured and the power density is obtained in accordance with Ohm's law. A peak power density of 239.13 mW/m^2 was achieved at a 500 MΩ load. Based on the excellent PAS1-TENG output performance, it was connected into a circuit with a rectifier bridge, capacitor and load (Fig. 5i) to assess its energy harvesting capability. The device under the contact-detach motion at 1 Hz successfully lighted up multiple LEDs and rapidly charged different capacitors (2.2 μF, 10 μF, 22 μF, and

33 μF) (Fig. 5j), with the charging rate decreasing as in capacitance increased. Furthermore, stored energy in a 33 μF capacitor effectively powered LEDs (Fig. 5k). These results indicate that the PAS1-TENG can serve as a self-power source for various electronic devices.

3.6. Self-powered sensing utilizing PAS1-TENG

To investigate whether PAS1-TENG work as a self-powered sensor for monitoring human motion, it is placed on the joints of fingers, elbows, knees, and wrists. It can be observed that PAS1-TENG effectively convert the mechanical energy from the motion of each joint into electrical energy (Fig. 6a and Fig. S14), showing a clear difference in output voltage. This indicates that it can be used as a self-powered sensor for monitoring human motion. In addition, a self-powered pressure sensor was created. To test the link between pressure and the output voltage of PAS1-TENG, various external forces were applied using a manometer, and the resulting output voltages were recorded (Fig. S15). It was found that PAS1-TENG responds in multistage stages to pressure, reaching a sensitivity (K) of 3.28 V/kPa at low pressure (0.5–4 kPa), while the output voltage remains stabilizes after the pressure exceeds 10 kPa

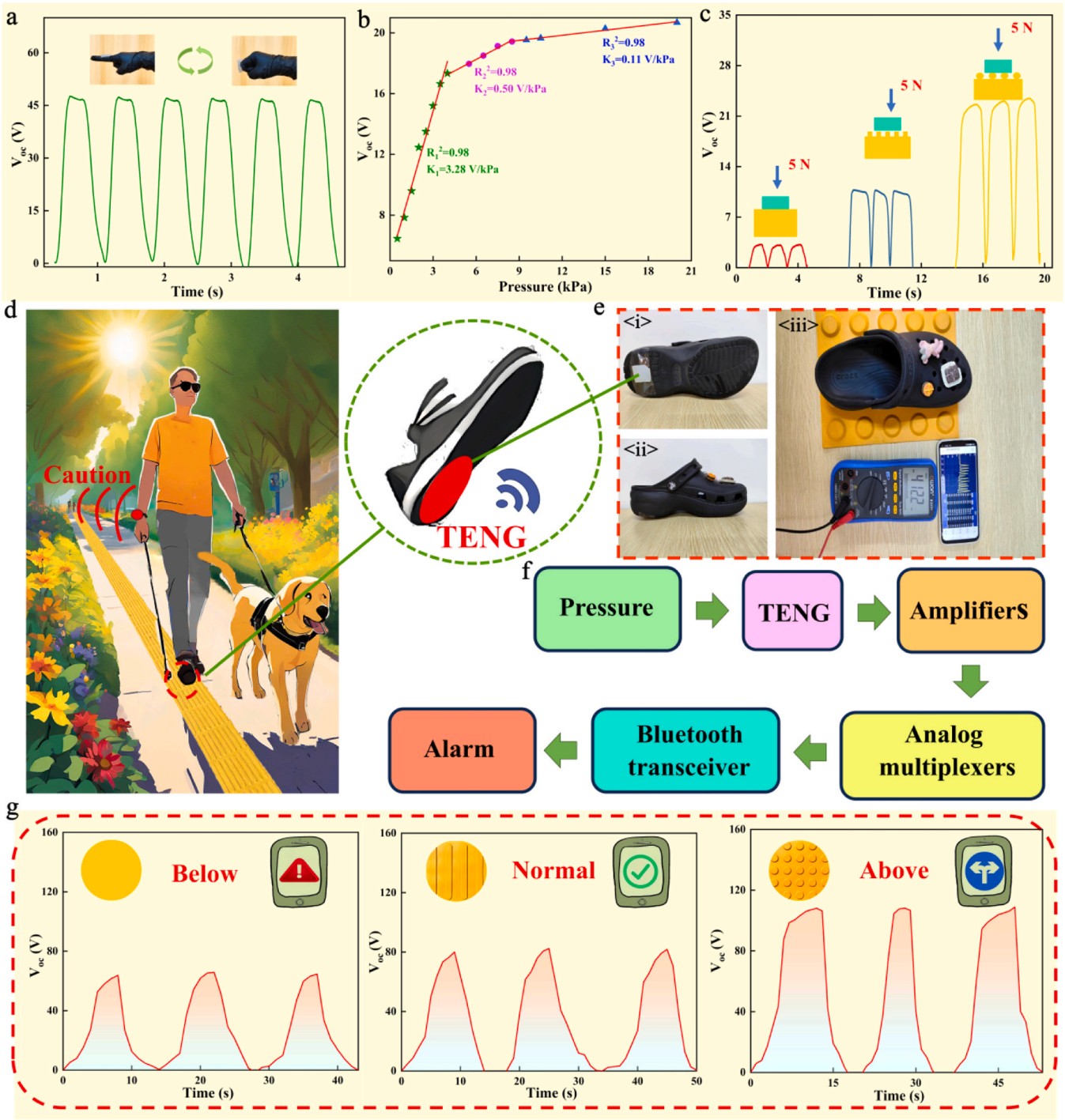


Fig. 6. a) V_{oc} generated by PAS1-TENG due to finger bending. b) K value of PAS1-TENG. c) Output response of PAS1-TENG for different textured pavements. d) Demonstration diagram of PAS1-TENG in blind traveling. e) <i-ii>Integration of PAS1-TENG into the sole of a shoe; <iii>Physical diagram of an intelligent pavement sensing system (IPS) based on PAS1-TENG. f) Working principle of IPS. g) Wireless detection of IPS.

(Fig. 6b). Notably, the response time is not affected, maintaining a quick response of 0.27 s (Fig. S16). There is also a significant difference output signal occurs when placing PAS1-TENG is placed on different textured pavements with certain force (Fig. 6c). Therefore, as shown in Fig. 6d-e, PAS1-TENG is fitted into the sole of a shoe and combined with a Bluetooth system to create an Intelligent Pavement Sensing System (IPS) aimed at helping blind individuals walk. Fig. 6f presents the working principle of the IPS system. When PAS1-TENG detects pressure, it generates a voltage signal that gets amplified by an amplifier, channeled though an analog multiplexer, and then sent to a cell phone to deliver corresponding warning via the Bluetooth module. As shown in Fig. 6g, the V_{oc} were about 60 V, 80 V, and 105 V when volunteers wore shoes with PAS1-TENG under normal ground, bar blind, and circular blind conditions. Furthermore, it was found that the walking speed does not affect the output voltage (Fig. S17). Therefore, 80 V is set as the threshold. If volunteers stray from the normal path, their mobile phone will pulse short and rapid vibrations to encourage the volunteers to adjust the direction. When they arrive at the circular blind path and need to change direction, the phone will give off long and gentle vibrations. The IPS system is expected to have a huge potential in the area of wireless smart navigation for the visually impaired.

4. Conclusion

In sum, multifunctional composite hydrogels with a double cross-linked network structure and nano-SiO_2 reinforcement were fabricated using silica-coated ZnO nanoparticles with unsaturated double bonds on the surface as core-shell structured particles. Under the multiple influences of SiO_2@ZnO, the hydrogel demonstrated excellent high strain, high adhesion, low hysteresis, and high electrical conductivity, among others. Therefore, a strain sensor based on the PAS hydrogel was developed, and its excellent sensitivity and fast response time ensured that it could accurately detect various human activities. In addition, a single-electrode triboelectric nanogenerator was assembled with PAS hydrogel serving as the conductive layer, which successfully charged and discharged various electronic devices. Additionally, an intelligent pavement sensing system was established in combination with a Bluetooth system to assist blind people in traveling. This paper demonstrates the promising development of PAS hydrogels in flexible electronic devices, and the designed PAS-TENG has significant potential for applications in wearable electronic devices, self-powered sensing, and other fields.

CRediT authorship contribution statement

Yuqi Li: Writing – review & editing, Supervision, Conceptualization. **Zhoujing Chen:** Writing – review & editing, Methodology. **Hong Ruan:** Writing – review & editing. **Xiaoli Liang:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.colsurfa.2025.137365.

Data availability

Data will be made available on request.

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